JORGE CHÁVEZ MARCELO GALLARDO

DYNAMICAL SYSTEMS AND OPTIMAL CONTROL IN ECONOMICS

Methods, Models and Applications

DEDICATION

To Juana, Manuel, and Elba,
three trajectories toward infinity.
To Rocío,
the initial condition.

J. Chávez

To Ana Sofía and Gonzalo,
may you keep growing at an exponential rate.

M. Gallardo

PREFACE

This book grew out of several years of teaching two courses, Mathematics for Economists and Mathematics for Control Theory, in the Departments of Economics and Mathematics at the Pontificia Universidad Católica del Perú. It is intended for undergraduate and graduate students in economics, applied mathematics, and engineering who have a working knowledge of calculus and linear algebra. It also serves as a reference for instructors and researchers interested in dynamic economic modeling and optimal control theory.

The book provides an accessible yet rigorous introduction to continuous-time dynamical systems and optimal control theory, with a strong emphasis on economic applications throughout. It includes 141 carefully structured exercises, distributed across the problem sets at the end of each section and often accompanied by hints, designed to guide learning, reinforce understanding, and assess student progress. This extensive set of exercises also constitutes a practical resource for instructors seeking comprehensive teaching materials.

A distinctive feature of this book is its systematic integration of economic models into every chapter. Each mathematical concept is developed alongside one or more economic applications, ensuring that the theory remains grounded in meaningful contexts. The range of applications spans classical topics, such as Malthusian and Verhulst population dynamics, Domar's investment model, and Walrasian price adjustment, as well as a broad collection of modern economic models. Among the dynamical systems applications, the book covers Solow's growth model, poverty trap dynamics,

the SIR epidemiological model, Goodwin's cyclical growth, and Kaldor's business cycle. As for optimal control, it treats the Ramsey–Cass–Koopmans model, the AK growth model, Romer's endogenous growth, Lucas's human capital accumulation, Tobin's Q, Hotelling's rule, Clark's renewable resource harvesting, the Nerlove–Arrow advertising model, and the principal–agent problem with infinite types. This last connection, between optimal control theory and the design of incentive-compatible contracts, illustrates how the mathematical tools developed here extend naturally not only to economic growth, but also to modern microeconomic theory. Many examples draw on seminal contributions, and several are inspired by recent publications, particularly in the fields of economic growth, development economics, and dynamic macroeconomics. This breadth allows mathematics students interested in economics to engage deeply with applied topics, while economics students seeking formal rigor will find the material both accessible and enriching.

There is a strong emphasis on visual intuition and graphical analysis, supported by abundant, carefully crafted figures created using `TikZ`, MATLAB, GeoGebra, and Wolfram Mathematica. Phase diagrams are prominently featured to facilitate a clear understanding of complex dynamic behaviors.

Our focus on continuous-time analysis allows for a coherent and self-contained exposition. Virtually the entire book is accessible to students with a basic background in vector calculus, real analysis, convex optimization, and linear algebra. Exceptions include Sections 3.4, 4.3 and the chapter appendices, which are appropriate for advanced undergraduate students or graduate-level

readers. Moreover, the inclusion of less-standard topics (such as bifurcation theory, Goodwin's cyclical model, limit cycles, modern economic growth models, and mechanism design) sets this book apart from others in the current literature and broadens its relevance for students and researchers interested in contemporary applications of dynamical systems to economics.

The book is organized into six chapters. Chapter 1 introduces dynamical systems through first-order scalar differential equations, covering fundamental solution methods (separation of variables and Bernoulli equations) alongside classic economic models such as Domar's investment model, Malthusian and Verhulst population dynamics, and Walrasian price adjustment. Chapter 2 extends the analysis to a general qualitative study of one-dimensional nonlinear systems, addressing Solow's growth model, bifurcation theory, a labor market model, poverty trap dynamics, and numerical methods including the Euler and Runge–Kutta schemes. Chapter 3 treats higher-dimensional linear systems, with particular emphasis on the two-dimensional case, which captures the core geometric intuition of the general theory; matrix exponentials and the Jordan canonical form are introduced as practical tools for obtaining explicit solutions, and the structure of stable and unstable subspaces is examined, laying the theoretical groundwork for the nonlinear analysis and optimal control applications that follow. Chapter 4 addresses higher-dimensional nonlinear systems through the Hartman–Grobman and Poincaré–Bendixson theorems, enabling the analysis of models such as species competition, Goodwin's cyclical growth model, Kaldor's business cycle, the SIR epidemiological model, and the

dual cross adjustment mechanism. Chapters 5 and 6 are devoted to dynamic optimization: Chapter 5 develops the calculus of variations, while Chapter 6 presents continuous-time optimal control theory and applies it to a broad range of economic models. Each chapter concludes with an appendix containing supplementary material (such as more involved proofs, additional theoretical background, or advanced topics) that, while not essential to the main exposition, enriches the presentation and provides further depth for interested readers.

This work was motivated by influential textbooks that have shaped the study of dynamical systems and economic modeling, including Seierstad and Sydsæter (1987), Léonard and Long (1992), Barro and Sala-i-Martin (2003), Gandolfo (2009), Perko (2013), Acemoglu (2009), and Sethi (2019).

We believe this book fills a gap in the current literature by offering a dedicated, coherent treatment of continuous-time dynamical systems and optimal control theory applied to economics. The continuous-time setting lets the material unfold as a single line of reasoning, with each chapter building on the last, from scalar equations to higher-dimensional systems and from the calculus of variations to optimal control. Throughout, a broad and up-to-date selection of economic applications ties every technique to a concrete economic question; the chapter appendices keep the exposition self-contained; and a sustained use of graphical tools, phase diagrams in particular, reinforces the geometric intuition behind the qualitative theory. The book suits courses on dynamic macroeconomics, mathematical methods in economics, applied

differential equations, and optimal control, in both one- and two-semester formats, and its accessibility, together with a rich set of exercises, makes it useful across academic levels.

Finally, we would like to thank the anonymous reviewers at Springer and the editorial team for their valuable suggestions and support throughout the preparation of this edition. This book was begun before the rise of generative artificial intelligence; only in preparing its final versions did we use the tools GPT-6 Thinking, Claude Opus 5.5, and Refine Ink, to help improve the exposition and to cross-check calculations. All results, arguments, and final wording were reviewed and verified by the authors, who are solely responsible for the content. We hope this book will prove valuable for undergraduate and graduate students in economics, applied mathematics, engineering, physics, and biology, as well as for instructors and researchers working in these areas.

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Symbols

The following is a brief list of the symbols which have been used in this book.

- $\mathbb{N}$: Set of natural numbers, $\mathbb{N} \triangleq \{1, 2, \dots\}$.
- $\mathbb{Z}$: Set of integers, $\mathbb{Z} \triangleq \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$.
- $\mathbb{Z}_0^+$: Set of nonnegative integers, $\mathbb{Z}_0^+ \triangleq \{0, 1, 2, 3, \dots\}$.
- $\mathbb{R}$: Set of real numbers.
- A^c : If A is a set, A^c denotes its complement.
- $A \subset B$: Set A is included in set B.
- $\mathbb{R}^n$: Euclidean space of dimension $n \in \mathbb{N}$.
- $\|\mathbf{x}\|$: If $\mathbf{x}$ is a vector of a vector space V, $\|\mathbf{x}\|$ denotes its norm.
- $\mathcal{B}(\mathbf{x}_0, \epsilon)$: Open ball with center $\mathbf{x}_0$ and radius $\epsilon > 0$.
- $\mathcal{N}_\mathbf{x}$: Neighborhood of point $\mathbf{x}$, i.e., a set containing an open ball centered at $\mathbf{x}$.
- $\sup A$: Supremum of the set A.
- $\preceq$, $\succeq$: Order relation. For a symmetric matrix Q: $Q \succeq 0$, positive semidefinite; $Q \succ 0$, positive definite; $Q \preceq 0$, negative semidefinite; $Q \prec 0$, negative definite.
- ∂A : Boundary of the set A.

- $x \cong y$: x almost equal to y.
- $\mathcal{M}_{n\times n}(\mathbb{R})$: Vector space of matrices of order $n \times n$, $n \in \mathbb{N}$, with real entries.
- $\mathrm{Ker}(A)$: Kernel of A.
- $|A| = \det(A)$: Determinant of A.
- $\mathrm{tr}(A)$: Trace of A.
- $\mathrm{Span}(A)$: If A is a set in a vector space V, $\mathrm{Span}(A)$ denotes the space generated by A.
- $C^n(A)$: Space of n times continuously differentiable functions on A.
- $\sigma(A)$: Set of eigenvalues of matrix A, also known as *spectrum*.
- $\mathbf{x}^\top$: Transpose of vector $\mathbf{x}$.
- $\mathrm{int}(X)$: Interior of a set X.

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